

Impact of Weather and Airport Volume on Flights

Final Report

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Abstract

This work explores the applicability of data analytics technologies to assess the impact of weather conditions and airport throughput on various flight characteristics in order to provide insights and solutions to a wide array of problems plaguing aviation. We construct an ETL pipeline using open-source data in order to build a data warehouse, supported by a star schema, that is subsequently leveraged to present OLAP data to the end-user, with the goal of enhancing the decision support process. Our analysis successfully addresses the multifaceted data warehousing requirements necessary to mitigate operational adversities in the aviation sector.

AI Usage Acknowledgment: We used Generative AI tools during this project. They helped us fix grammar, write code, build database queries, and create Pentaho workflows and Tableau charts.

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1 Introduction

Despite advancements in logistics and predictive technology, transportation hubs worldwide remain susceptible to many volatile factors, such as environmental disruptions and structural constraints, that lead to overall systemic inefficiency. In aviation, these problems can be exacerbated, due to the complex boarding processes present at most airports and dependency on favorable atmospheric conditions to provide safe and timely services. As such, it is imperative to analyze flight data in order to extract useful insights about predominant factors impacting aviation throughput, passenger behavior, and ticket pricing, so that both airport management and customer can be improved.

To evaluate the aforementioned adversities and their subsequent impacts on aviation, we employ data analytics methodologies commonly found across various data-driven solutions. More specifically, we explore and collect flight, weather, and passenger-related records across US airports, apply Extract, Transform, Load (ETL) principles to store the collected data in a Data Warehouse (DW), and utilize the resulting database for querying and analysis through Online Analytical Processing (OLAP) and Data Mining (DM) in order to answer questions like:

1. Does extreme weather significantly impact travel time, flight cancellations, or delays in specific geographical locations? (DW / Diagnostic)
2. Are flight delays related to the daily passenger volumes recorded at TSA checkpoints? (DW / Diagnostic)
3. Does the number of flights or the geographical location of a city influence the average ticket price? (DW / Descriptive)
4. How do ticket prices vary between "stormy" and "clear" quarters, and do high ticket prices influence the rate of cancellations? (DW / Descriptive and Diagnostic)
5. Based on historical weather patterns and city travel numbers, what are the predicted ticket prices for the next quarter? (DM / Predictive)
6. Which months and cities are recommended for travel to minimize the risk of weather related delays and high costs? (DM / Prescriptive)

2 ETL Process

In this section, we document the various steps and decisions taken across the ETL pipeline, ranging from the data sources considered for the project to the pre-processing and storage of the extracted data.

2.1 Data Sources

To answer our questions regarding the impact of weather conditions and passenger volume on flight services, we started by identifying four core components that are fundamental to our study: detailed flight records, weather data regarding each flight, passenger volume at airports, and flight ticket prices. Our hope here is to analyze how the weather and passenger data related to a flight's airports influence its trip delay, potential cancelation, and ticket price. To this end, we identified a collection of open source datasets from some US governmental institutions that could aid in our project:

- **BTS Flight and Market Statistics¹:** The Bureau of Transportation Statistics (BTS) contains detailed monthly records of domestic flights, with key features such as flight date,

¹<https://www.bts.gov/>

duration, airline, origin, destination, actual and estimated departure and arrival times, among others. Additionally, the BTS also provides quarterly market data, with details such as the average domestic airline fares and daily passenger volumes for various airports in the U.S. territory.

- **NOAA Daily Summaries²:** The National Oceanic and Atmospheric Administration (NOAA) is the US’s governmental agency responsible for weather monitoring and forecasting. They host public databases containing meteorological data, such as precipitation, minimum and maximum temperatures, and snowfall, recorded daily at thousands of weather stations worldwide;

Although we were able to find a substantial amount of data covering the time interval between 2022 and 2025, two limitations/challenges were immediately apparent:

1. A primary challenge was the lack of a direct relational link between flight records and meteorological data. This was mitigated by performing a spatial join, using latitude and longitude coordinates to map each airport to the nearest functional weather station.
2. The analysis of ticket pricing and passenger volume is constrained by the quarterly nature of the source data. This lower temporal resolution limits the depth of DM insights and restricts the granularity of certain DW queries.

2.2 Data Transformation

In order to pre-process the data to submit it to the DW, we relied almost entirely on Pentaho, by constructing adequate transformations for each data table and subsequently aggregating and managing them through jobs, with some minor Python interventions to facilitate integration with Pentaho. These interventions were solely dedicated to addressing the problem mentioned in Section 2.1, where two new columns were added to each flight record to indicate the nearest weather station to both the departure and arrival airports.

Focusing on the Pentaho pipeline, we started by developing three data cleaning transformations for the datasets extracted from the sources in Section 2.1:

- **Weather Cleaning:** The weather cleaning step (Figure 1) receives four datasets, each containing a year’s worth of daily precipitation, temperature, and snowfall measurements from various weather stations across the world. The transformation filters out measurements from non-US stations and keeps only precipitation, temperature, and snowfall values that are not missing. It then removes unnecessary columns (flags) and denormalizes rows in order to facilitate search by weather station and date.

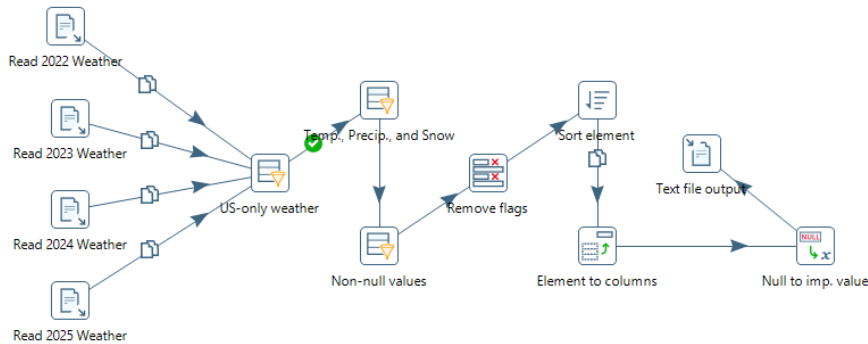


Figure 1: Weather cleaning transformation

²<https://www.ncei.noaa.gov/>

- **Flights Cleaning:** The flights cleaning step (Figure 2) starts by excluding entries with missing coordinates, replaces attribute names and codes for readability purposes, cleans missing values, splits hours, minutes, and dates between separate columns, and eliminates unnecessary columns.

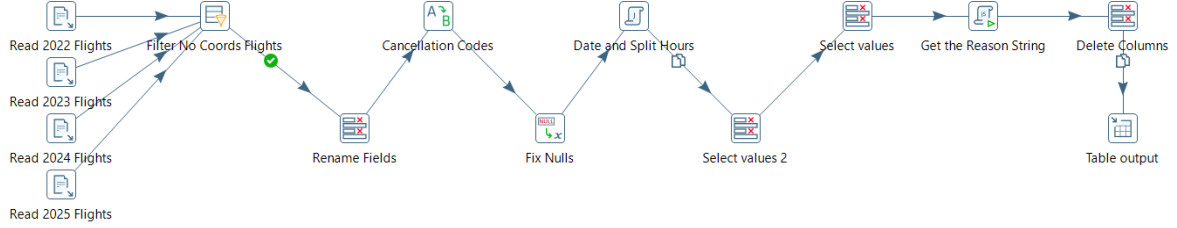


Figure 2: Flights cleaning transformation

- **Market Traffic Cleaning:** Market cleaning (Figure 3) filters the extracted dataset for the 2022–2025 period and the respective quarters. It selects key variables, such as City Market IDs, fares, passengers, and date fields, while defining data types and removing unnecessary columns. The refined results are then stored in a final temporary table.



Figure 3: Market cleaning transformation

2.3 Data Loading

To store the transformed data, we utilized a PostgreSQL database, hosted through a virtual machine in the DEI network. The overall relational schema for the database was based on the Star Schema in Figure 5 and created using ONDA³.

Before pre-processing, we estimated the aggregated size of all files to be sent to the database to be around 11GB. After the data cleaning process and indexation of multiple columns across every data table, we ended up with the six tables specified in the Star Schema, each one detailed in Table 1 and totaling around 3,556.7MB, a reduction of almost 68% over the initial dataset collection.

Another improvement we made refers to the time taken to execute all transformations and load the data to PostgreSQL. In our first attempt to insert data into the database, we constructed a pipeline that operated only within Pentaho. However, the time taken to upload only a year’s worth of data was over 8 hours (32+ hours for all the data), which we found to be too long. After running some tests, we identified that the indexing transformations were a critical bottleneck within the pipeline. To solve this, we decided to create temporary tables in the PostgreSQL database to solve the indices via querying. We registered an improvement of around 98% as this time we were able to upload all the transformed data within 43 minutes, as opposed to the over 30 hour estimate we had on our first run.

³<https://onda.dei.uc.pt/v4/>

Table 1: PostgreSQL Table Statistics (Project Schema)

Schema	Table	Estimated Rows	Data Size	Index Size	Total Size
project	flight	20,360,852	2744 MB	436 MB	3180 MB
project	time_	1,929,157	172 MB	41 MB	213 MB
project	weather	1,718,053	126 MB	37 MB	163 MB
project	market_traffic	52,420	3952 kB	1280 kB	5232 kB
project	delay	22,225	1528 kB	504 kB	2032 kB
project	airport	196	48 kB	16 kB	64 kB

2.4 Data Integration

To automate the data pipeline, we established a structured ETL (Extract, Transform, Load) workflow (Figure 4). While the initial logic was modeled using Pentaho jobs to ensure a logical processing sequence, the final implementation relied on optimized SQL queries within PostgreSQL to achieve the 98% performance improvement detailed in the previous section.

The integration process follows a strict dependency order, beginning with the independent dimension tables: Weather, Airport, Time, and Delay. For these tables, the logic involves extracting relevant attributes from the cleaned datasets, followed by a sorting and deduplication process to ensure data integrity. These refined records are then inserted in bulk into the database. A clear example of this is the Airport dimension, because raw flight data lists airports as both origins and destinations, we extract both sets of coordinates and metadata, merge them into a unified list, and remove duplicates to create a master reference of all unique locations.

The dependent fact tables required a more complex integration to bridge disparate datasets. For the Market Traffic table, we utilized temporary staging tables to execute relational joins between flight records and market data. This SQL-driven logic involves four key steps:

1. Mapping routes by matching origin and destination city IDs between the flight and market datasets;
2. Deduplicating the results to ensure that only one record exists per route per quarter, matching the granularity of the reporting;
3. Extracting essential metrics, including average fares, passenger volumes, and temporal markers (year and quarter);
4. Organizing the results by Airport IDs to streamline the subsequent bidirectional insertion, which ensures data is available for both City A $\rightarrow$ City B and City B $\rightarrow$ City A directions.

Finally, the Flight fact table is populated through a series of simultaneous lookups that link each individual flight to its specific environment. This includes a Weather Correlation, which retrieves the condition ID based on the flight date and the station nearest to the airport; Temporal Links, which connect the flight to multiple timestamps in the Time dimension (such as estimated vs. actual times); and Operational Details, which link the record to the Delay dimension. By resolving these foreign keys via SQL queries rather than traditional row-by-row lookups, we were able to commit the complete flight records to the final fact table with significantly higher efficiency.

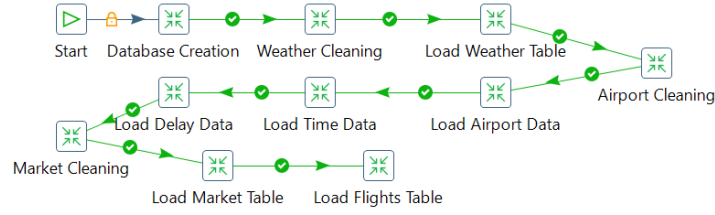


Figure 4: Pentaho Workflow

3 Data Warehouse Design

3.1 Star Model

The star schema (Figure 5) is centered around two primary fact tables, Flight and Market Traffic, which allow for the analysis of both operational and economic data. The Flight fact table operates at an individual flight basis, capturing specific metrics such as distance, estimated and actual durations, and status indicators for diversions or cancellations. The Market Traffic table tracks broader market trends, specifically quarterly averaged fares and daily passenger volumes. These fact tables are supported by four main dimension tables:

1. **Airport**, which provides geographic context such as city, state, and coordinates;
2. **Weather**, which records precipitation, temperature, and snowfall;
3. **Time**, which offers high granularity from the quarterly level down to the minute;
4. **Delay**, which categorizes the specific reasons and durations of flight disruptions.

The design choices for this schema are directly driven by the research questions mentioned previously. By linking the Weather and Airport dimensions to the Flight fact table, the model can identify if extreme weather causes flight cancellations or trajectory disruptions in specific locations. The inclusion of the Market Traffic table allows for the correlation of operational performance with financial data, enabling an analysis of how passenger volumes and weather conditions influence ticket pricing. Finally, the detailed Time dimension supports the transition from descriptive analysis to predictive modeling, providing the historical temporal patterns necessary to forecast future prices and recommend optimal travel months to minimize costs and risks.

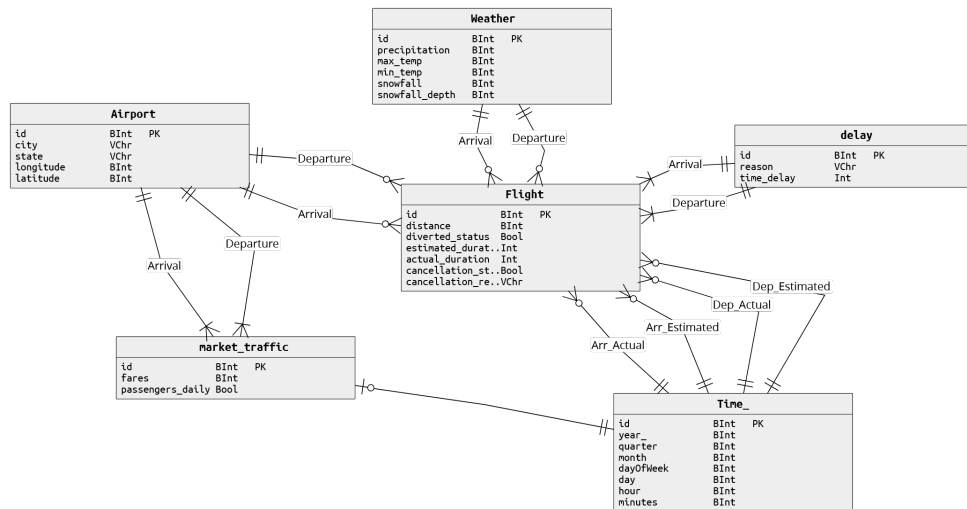


Figure 5: Star schema diagram

3.2 Answering the Big Questions

To answer the proposed requirements, the data warehouse connects operational flight data with environmental and economic dimensions. Here is how the schema addresses each specific question:

- **Question 1 (Extreme Weather Impact):** This is answered by joining the Flight fact table with the Weather and Airport dimensions. By filtering for precipitation or snow-fall extremes in the Weather table, we can measure how these conditions correlate with cancellations and flight durations/delays;
- **Question 2 (Passenger Volume and Delays):** This requirement utilizes a join between the Flight fact and the Market Traffic, via shared Time and Airport keys. This allows us to compare daily passenger counts against the delay durations recorded in the Delay dimension;
- **Question 3 (Geography and Ticket Prices):** This is addressed using the Market Traffic table and the Airport dimension. We can analyze if the city location influences the average fares for specific routes;
- **Question 4 (Stormy Quarters and Cancellations):** This uses the Flight and Market traffic tables joined with Time (filtered by quarter) and Weather. It identifies stormy periods to see if ticket prices in those quarters correlate with higher cancellation rates in the Flight table;
- **Question 5 (Price Prediction):** This predictive task relies on the historical relationship between fares and time. By looking at seasonal and environmental patterns over 2022-2025, the model uses these dimensions to forecast next-quarter pricing;
- **Question 6 (Travel Recommendations):** This prescriptive analysis aggregates data across the Flight, Delay, and Market Traffic tables. It filters by the Time and Airport dimensions to identify specific cities that offer the lowest average costs and shortest delay durations.

4 OLAP Presentation and Analysis

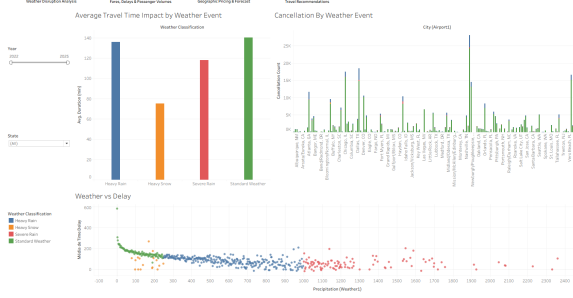
This section delves into the various visualizations that constitute the dashboard created to facilitate the user's data navigation and decision support processes.

4.1 User Access and Interface

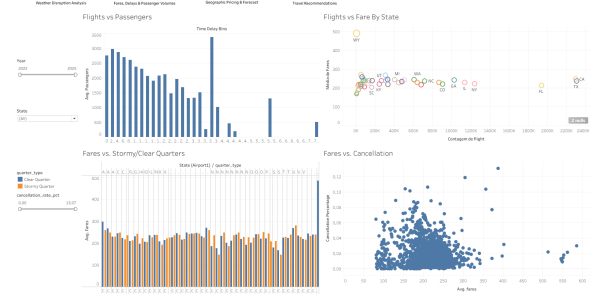
Through the use of Tableau to aid in the presentation of OLAP data to the final user, we developed an interactive, 4-tab dashboard (Figure 6). Each of these tabs encompasses a distinct category of visualizations specifically designed to address the research questions established in Section 1:

1. **Weather Disruption Analysis:** In this tab (Figure 6a), the user is able to check how weather conditions impact different aspects of a flight, such as travel time, cancellations, and delays across different years and locations in the US.
2. **Fares, Delays & Passenger Volumes:** Here (Figure 6b), the user might get insight into how fares correlate with location, flight cancellations, or weather, and determine if passenger volume influences flight delays, according to the year, location, and cancellation percentage.

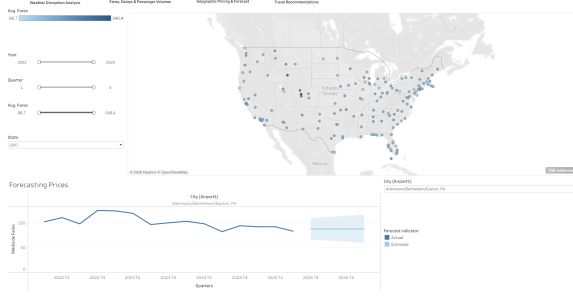
3. **Geographic Pricing & Forecast:** This tab (Figure 6c) shifts the focus to a more geographical standpoint, where the user can study if geographical location impacts ticket pricing, and utilize a tool to forecast future fares.
4. **Travel Recommendations:** In this last tab (Figure 6d), the user can discover the best time and places to travel based on average fares and flight delays for a given time and location.



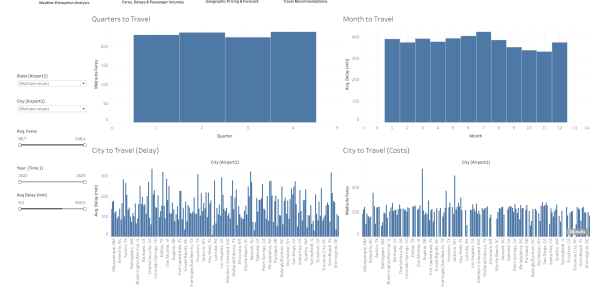
(a) Weather Disruption Analysis



(b) Fares, Delays & Passenger Volumes



(c) Geographic Pricing & Forecast



(d) Travel Recommendations

Figure 6: Comparison of the four primary dashboard tabs used in the analysis.

4.2 Analyses Performed

The analytical framework implemented in our Tableau dashboard covers four distinct types of analytics to provide a comprehensive view of the aviation landscape.

- **Descriptive Analysis:** We used bar charts and geographical scattermaps to summarize ticket price data under various conditions. For example, the "Travel Recommendations" describes which months and quarters have higher fares or delays, allowing users to understand baseline trends in the industry;
- **Diagnostic Analysis:** To understand why disruptions occur, we analyzed the relationship between weather events and operational metrics (Figure 6a). By correlating specific weather classifications with cancellation counts and average delay times, we can diagnose how environmental impact affects the airport systems. Similar correlations were studied for flight delays vs. daily passenger volumes and ticket prices vs. cancellations (Figure 6b).
- **Predictive Analysis:** To analyze future ticket prices, we applied a time-series forecasting to predict future ticket prices (Figure 6c);
- **Prescriptive Analysis:** The dashboard offers recommendations by identifying optimal travel windows (Figure 6d). By filtering for the lowest fare costs and shortest delays, the system prescribes the best times and locations for travel to minimize risks and costs.

4.3 Findings and Discussion

The analysis of the aviation landscape through our data warehouse provided high-level insights, though the scale of over 20 million records made it complex to isolate every local behavior. While "Standard Weather" accounts for the vast majority of flight volume, rare events like "Heavy Rain" and "Severe Rain" cause disproportionate spikes in travel time and delays. Because most flights occur in normal conditions, these disruptions are often localized and more noticeable in specific geographical hubs than in national averages.

We also observed a complex relationship between pricing and performance. While "stormy" quarters often correlate with higher fares, these prices do not consistently lead to higher cancellation rates, suggesting that market demand remains resilient during operational friction.

Predictive modeling for ticket prices remains a significant challenge because of the limited temporal resolution of the market data. While the total dataset is massive, the ticket prices and passenger volumes are only reported as quarterly averages. This means that for any specific city, we essentially have a maximum of four data points per year. Across the 2022–2025 study period, this results in just 16 price observations per location to feed the prediction model. This lack of granular, high-frequency pricing data restricts the depth of insights and makes it difficult to forecast future fares with high precision.

5 Conclusions

This project resulted in a robust data warehouse architecture designed to analyze the complex relationships between flight operations, weather and market economics. While we technically addressed our research questions, the findings often remained high-level due to "big data bias", as the vast majority of standard flights overshadowed rarer disruption events.

A primary lesson learned was that managing over 20 million records requires significantly more technical expertise and investigation than initially anticipated. The massive volume and low quarterly resolution of market data frequently obscured localized nuances, demonstrating that data at this scale demands sophisticated statistical controls to isolate meaningful insights from environmental noise.

Moving forward, future work will focus on refining the star schema by utilizing granular temporal attributes, such as hour and minutes timestamps, which are currently not used. We could also plan to investigate the impact of specific airport carriers and state-level behaviors to overcome broad data bias and provide more prescriptive insights for aviation stakeholders.